Class 4: Bed Configurations (Part 2)



John Southard
Emeritus MIT

What a strange, strange boy

He sees the cars as sets of waves
Sequences of mass and space

Joni Mitchell,

'A strange Boy'

Hejira, Asylum Records

Class outline: How do water flows interact with a bed made of sand?

- Describe the bedforms that develop in fine-coarse sands under unidirectional flows (dunes to antidunes)
- Why ripples and dunes migrate downstream

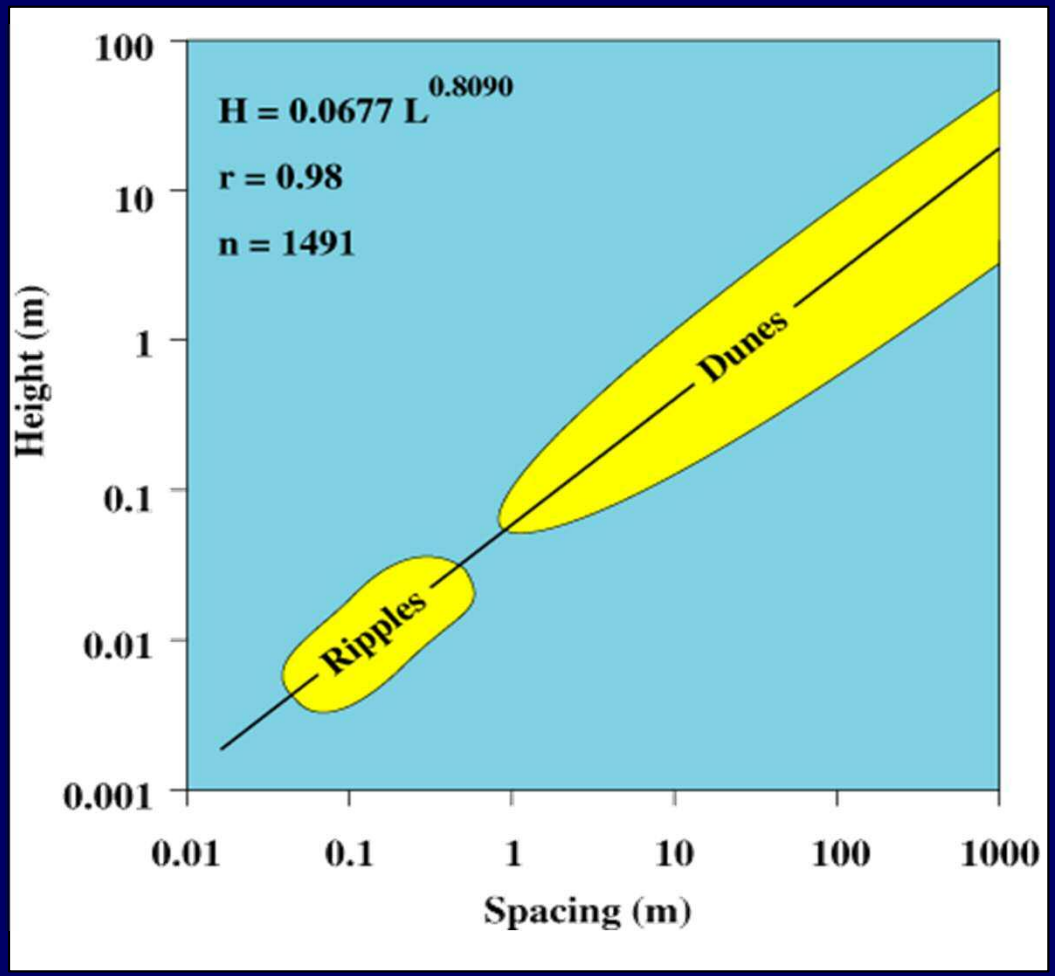
Dunes

Large, asymmetric bedforms
Not 'large ripples' – ripples don't grow into dunes- transition is abrupt

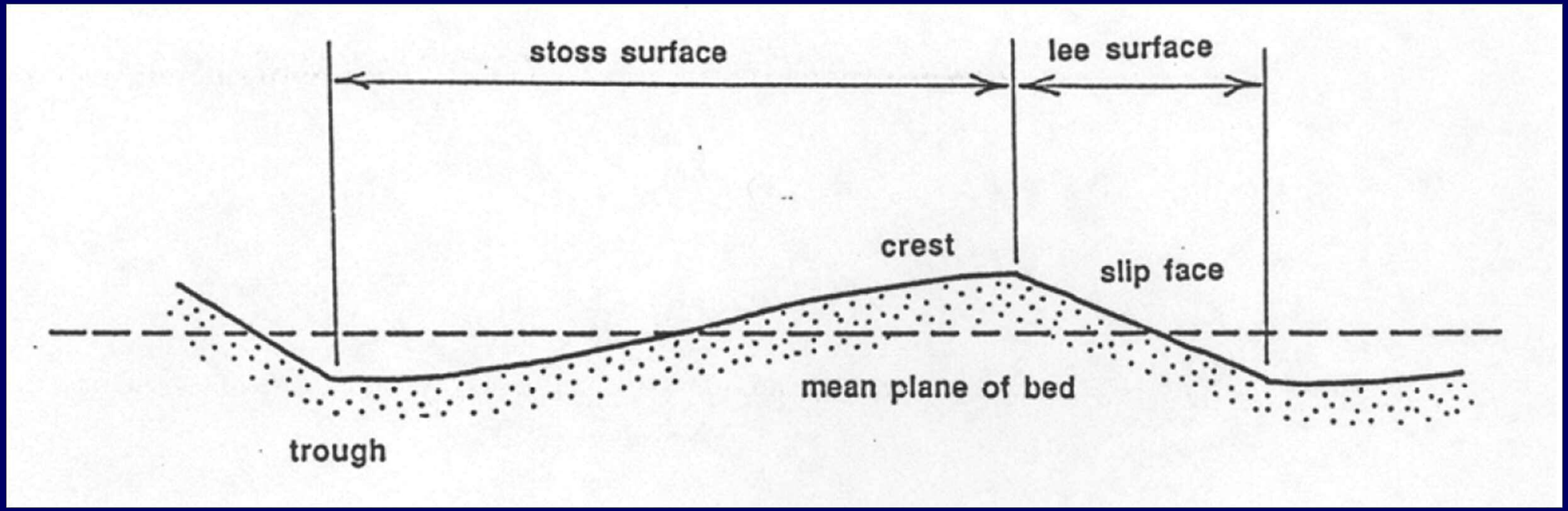
- Never first (lowest velocity) bedform to appear
- don't form in sediment <~0.2 mm; require hydrodynamically rough boundary
- most common bedform in sand bed rivers; also occur in gravel bed rivers



Dunes in the North Loup River, Nebraska USA.
Two people are circled for scale.
Image courtesy D. Mohrig.

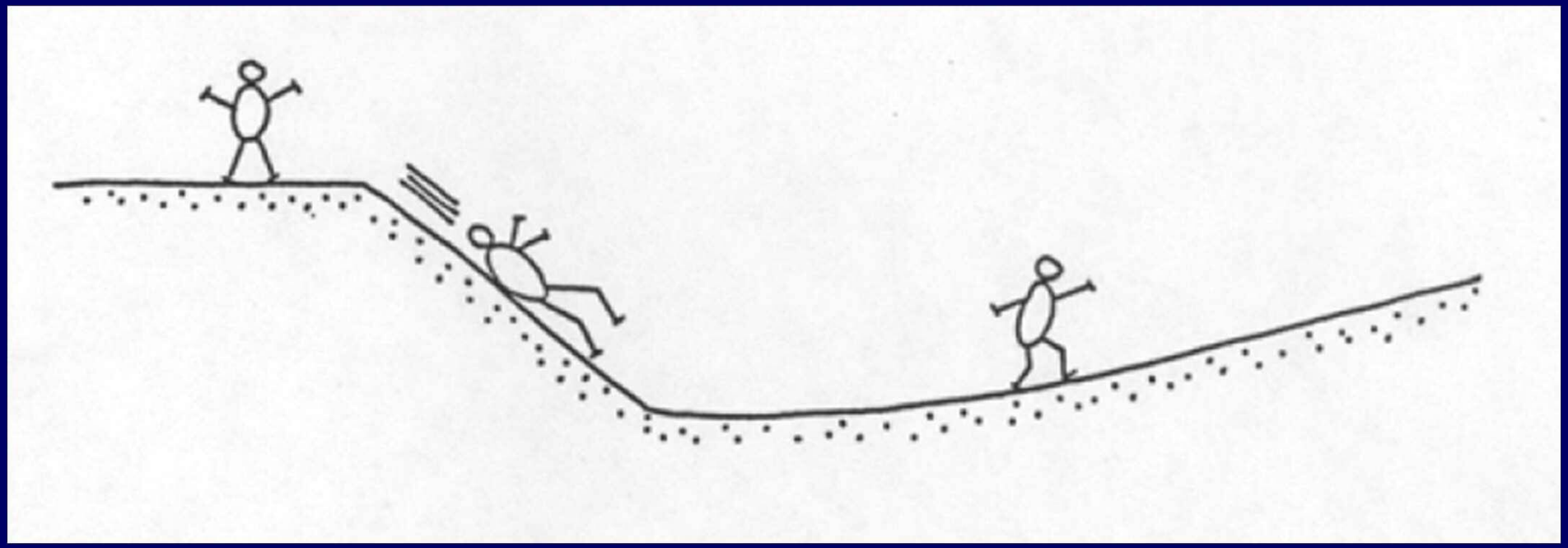


Literacy: geometry of ripples and dunes



Gentle stoss (upstream) side and a steep lee (downstream) side

Observation: Take a tour over a ripple or dune profile

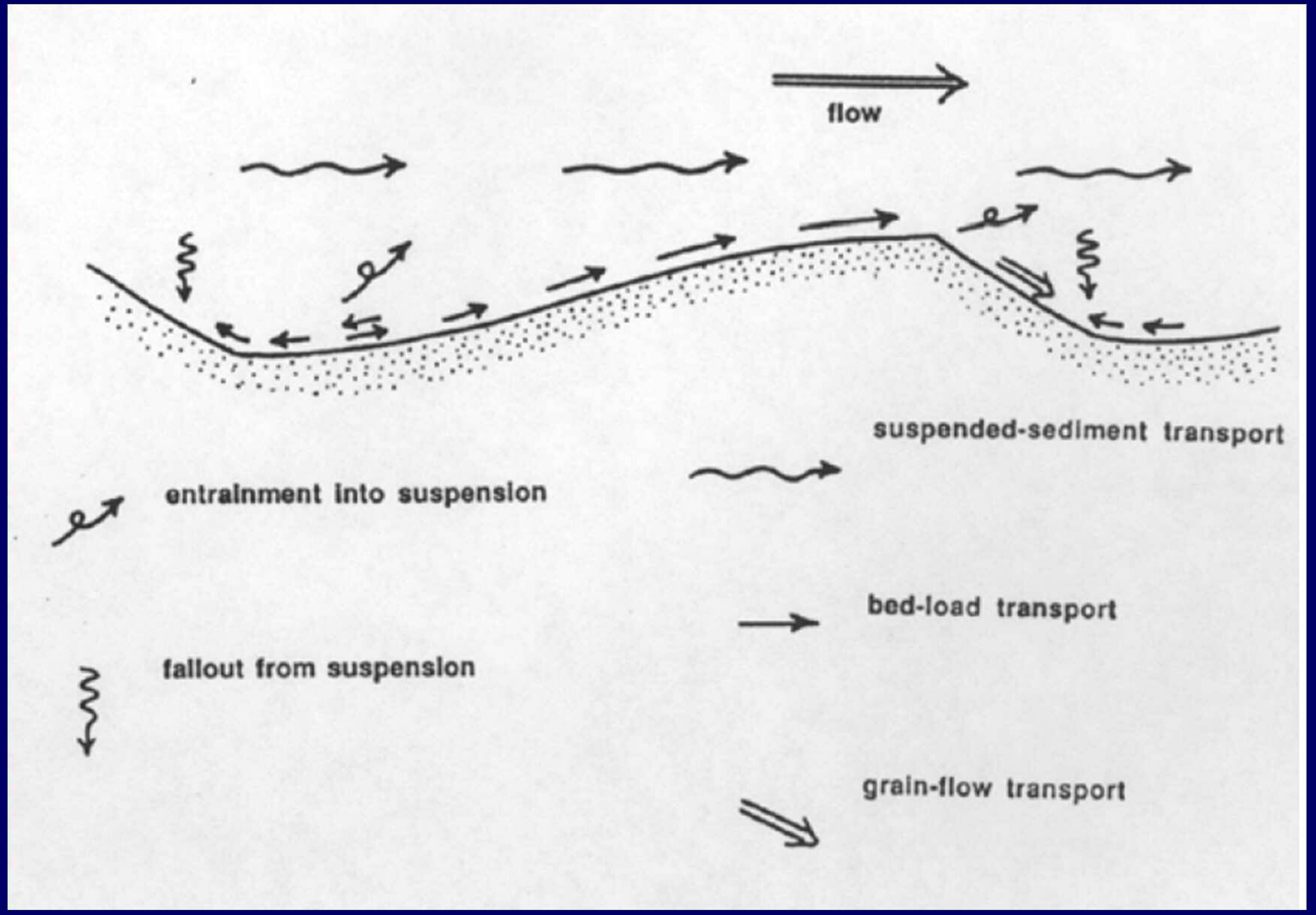


Starting at a crest, sliding down the slip face, and then walking across the trough and up the stoss surface downstream

The flow you'd sense would differ greatly along the profile.

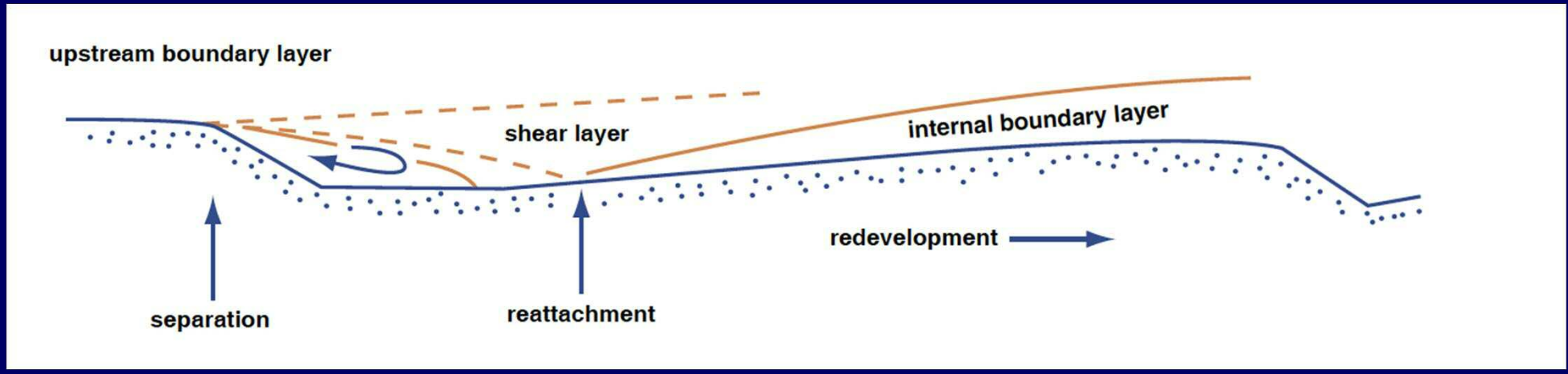
You could actually do this on a big dune in a river or a tidal current, or on a subaerial dune

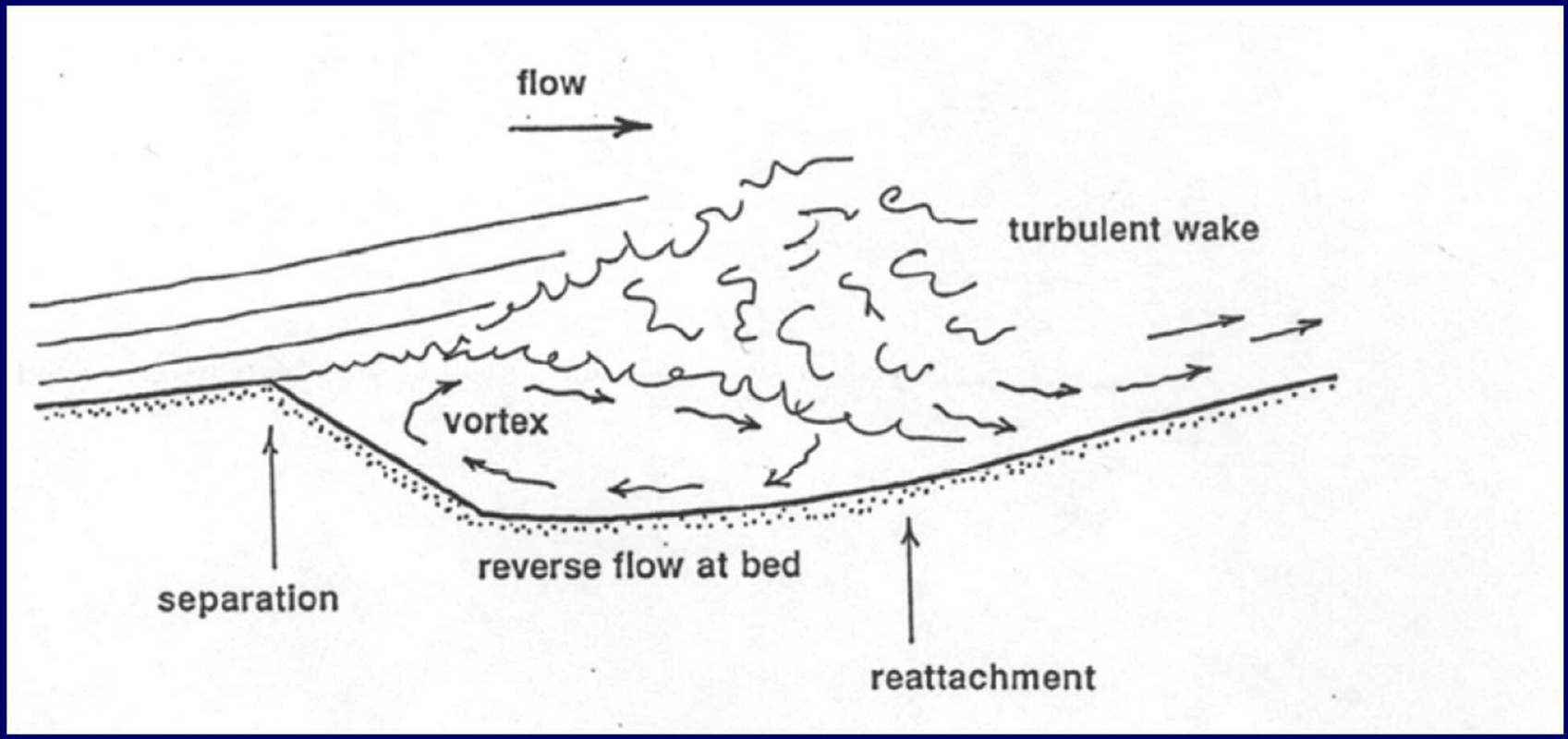
Observation: sediment movement over ripples and dunes



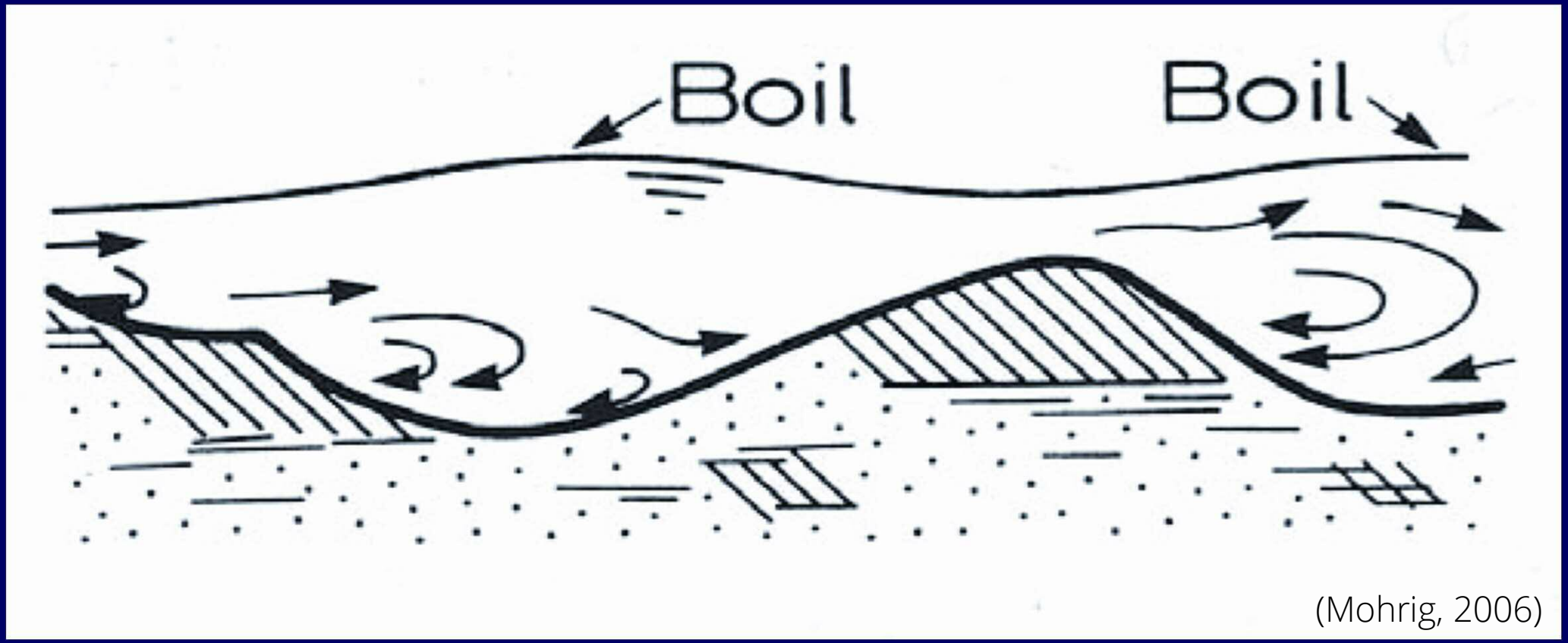
Note changes in the way sediment is transported over the dune

Observation: flow over ripples and dunes





Observation: Interaction between dunes and the water surface



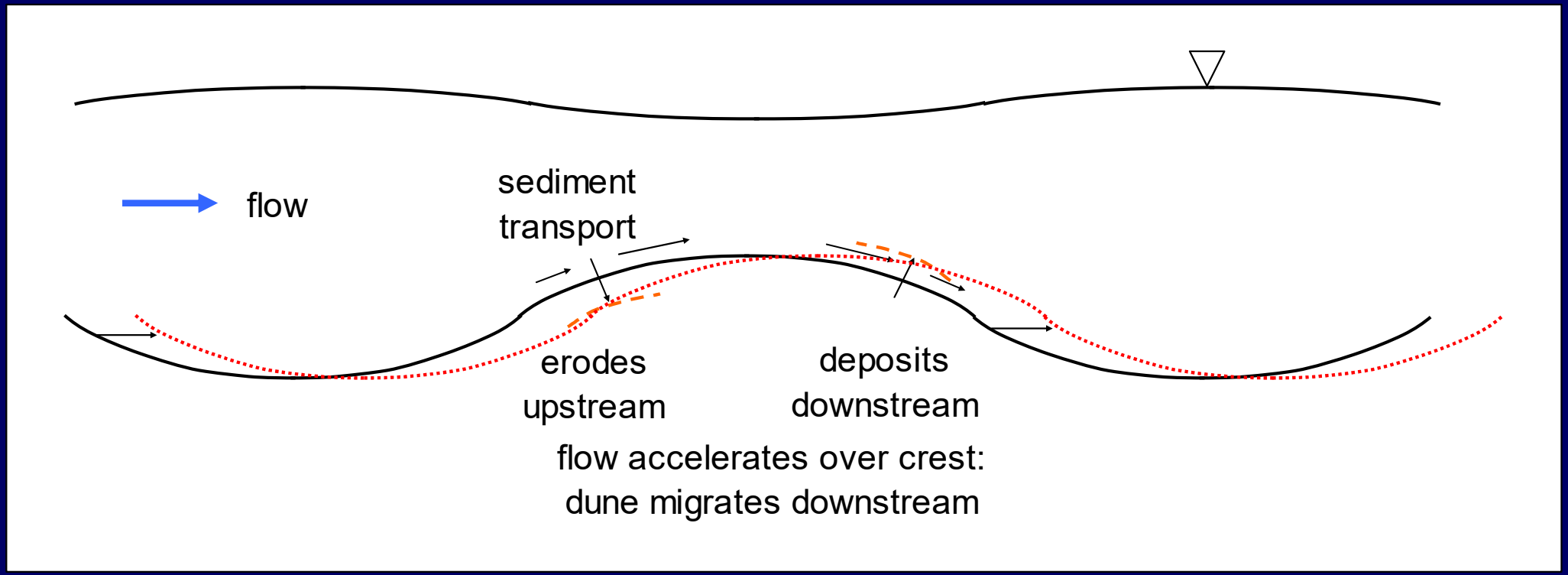
(Mohrig, 2006)

Bursting cycle resulting from dune topography

Observation: Process by which ripples and dunes migrate

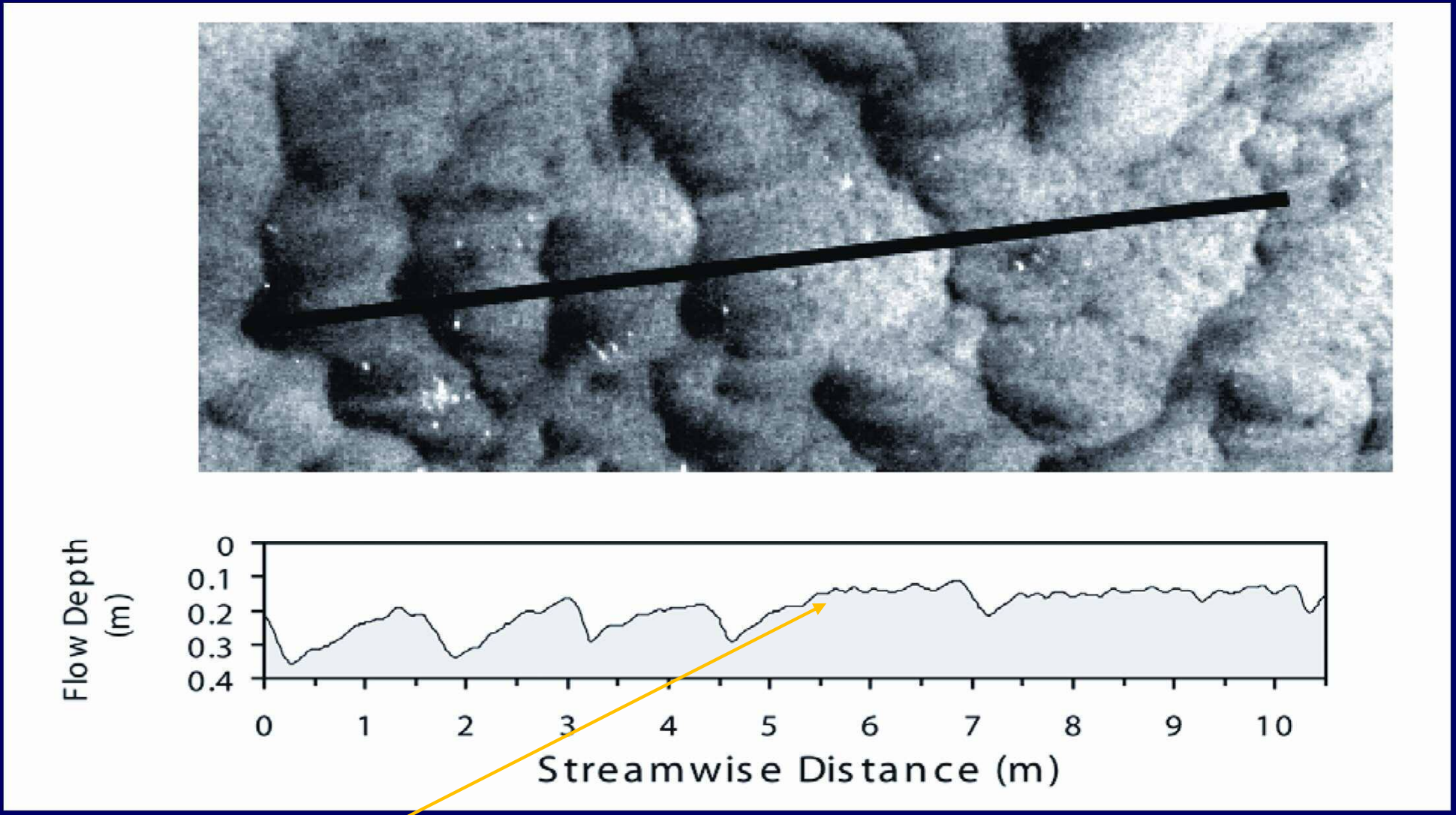
Water surface is out of phase with the bed

- Depth variation is out of phase with the bed
  - Flow accelerates from trough to crest
  - Sediment transport increases from trough to crest
  - Bedform migrates downstream
  - Bedform becomes asymmetric



Differences between Ripples and Dunes

Ripples are sometimes superimposed on dunes in equilibrium to form compound bedforms



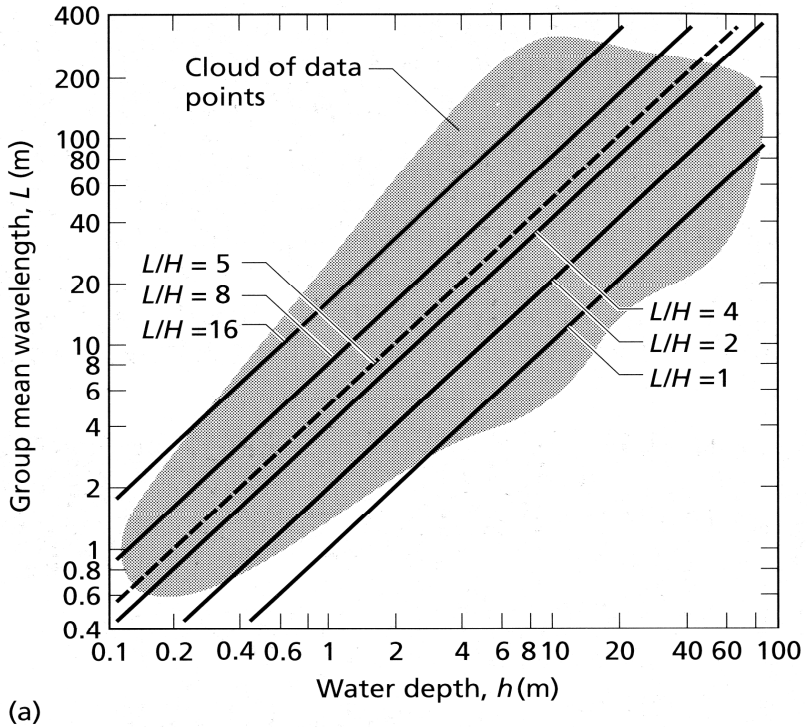
ripples superimposed on stoss side of dunes

(Mohrig, 2006)

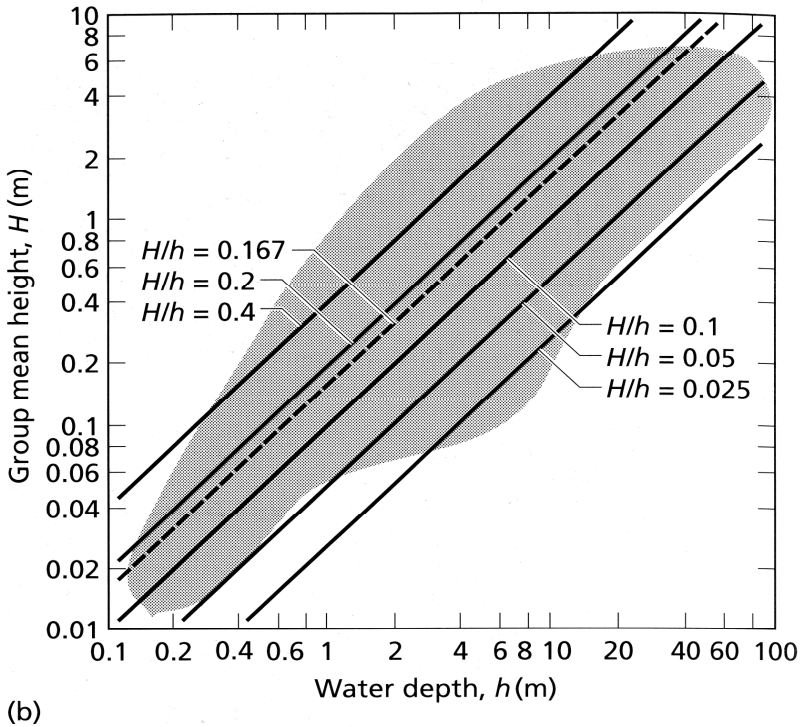
Constraints on dune dimensions: dune height

Positive relationship between dune height and flow depth

- (0.2-0.3 times vertical scale of flow)



(a)

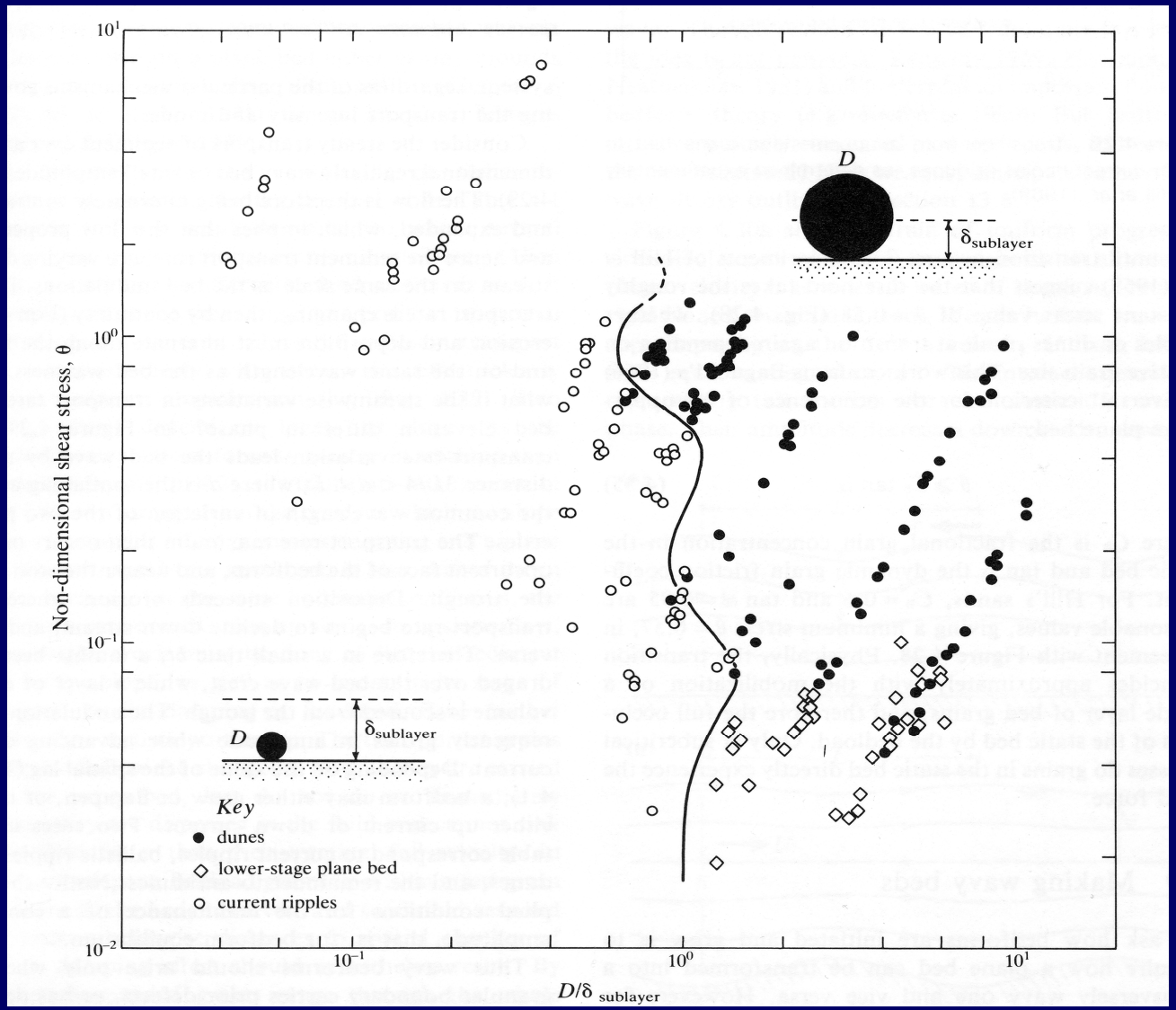


(b)

Fig. 7.9 Graphs to show the correlations between (a) dune group mean wavelength and mean water depth and (b) dune group mean height and mean water depth, both for one-way river flow only. (Data assembled by Allen, 1982.) Despite the considerable scatter of data (generalized here), it is apparent that the scale of dunes increases with boundary-layer thickness, implying, but not proving, that dune bedforms owe their origins to flow processes that affect the whole outer boundary layer.

(Leeder, 1999)

Why do dunes scale with the flow depth when ripples do not?



Experimental occurrence of current ripples, dunes and lower stage plane beds in quartz sands and silts, as a function of non-dimensional bed shear stress and relative grain size (ratio between grain size (D) and thickness of the viscous sublayer) (Allen, 1985)

Dunes: just how big can they get?



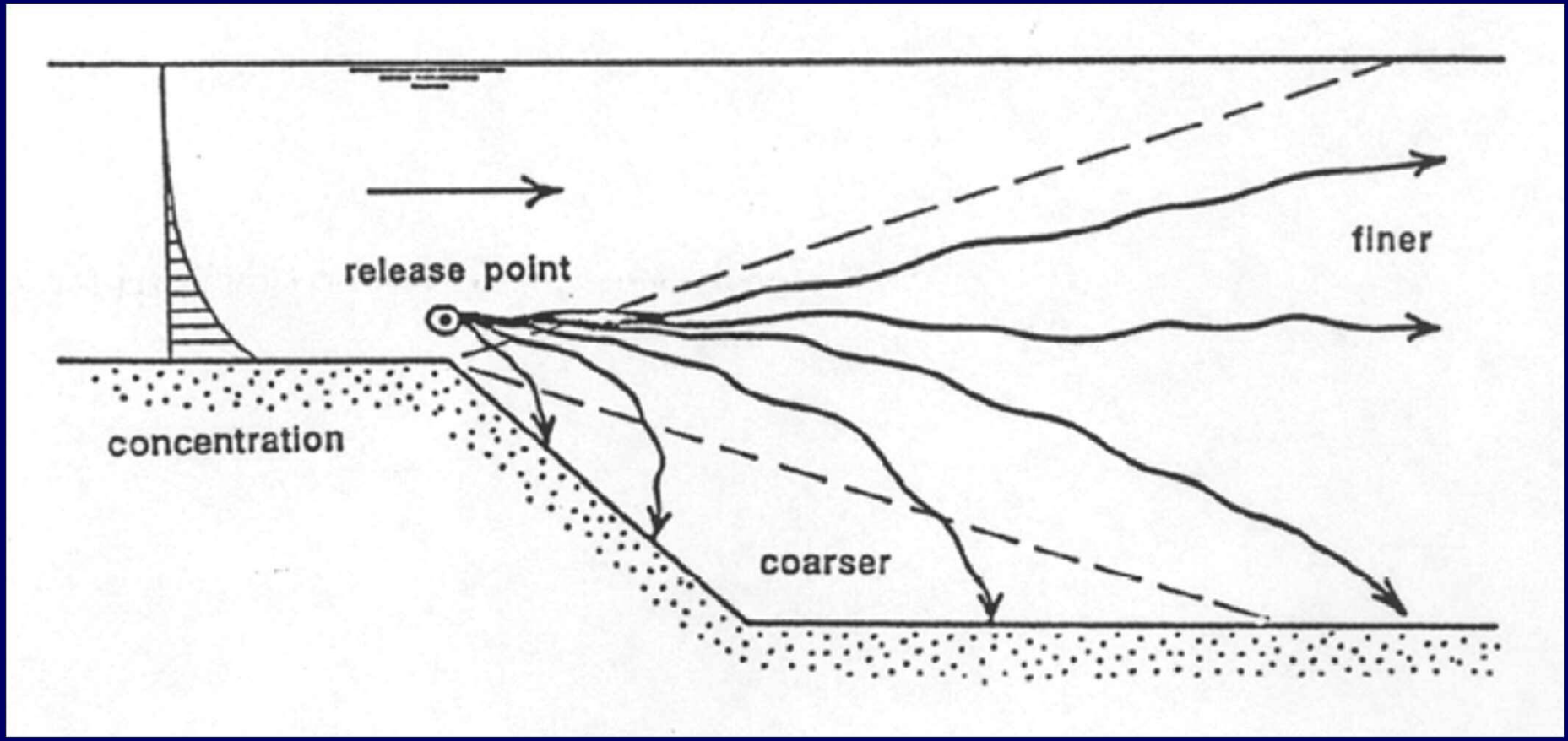
Superimposed aeolian bedforms

Dune planform varies with flow strength

At lower velocities:
- dunes low, widely spaced and 2-D
- long, straight to sinuous crests
- lee slope near angle of repose (25°- 30°)
- min. height 0.05 m
- form ratio 50-200
- sediment transport dominated by rolling, sliding and avalanching

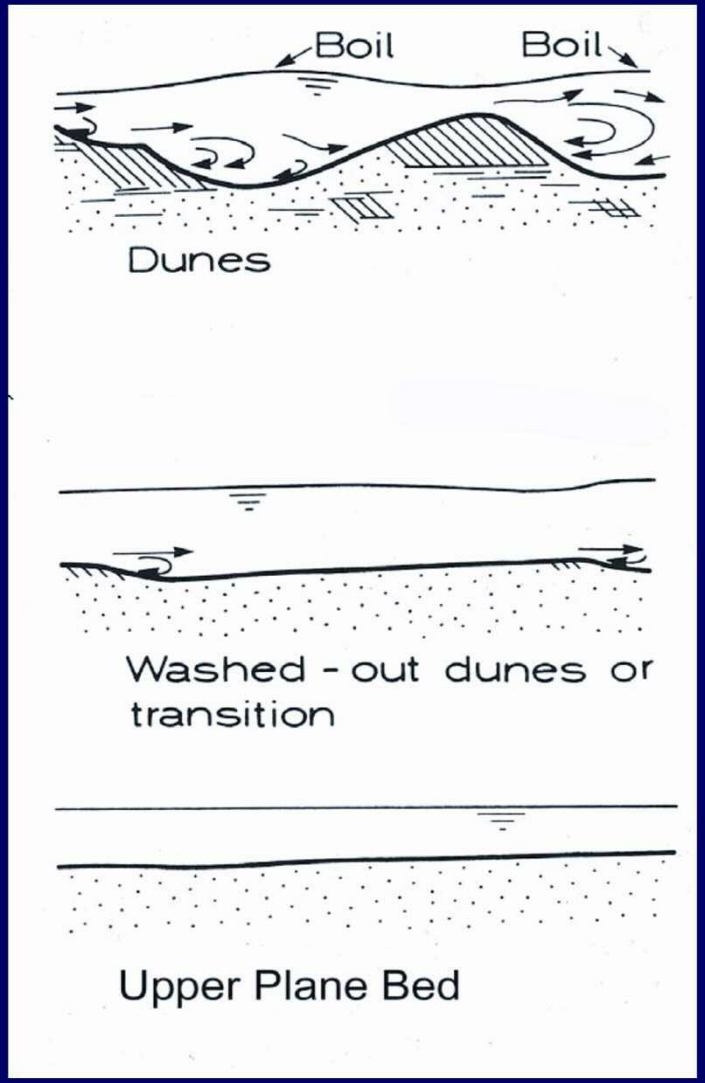
As velocities increase gradual transition to 3-D dunes:
- height increases without increase in spacing
- complex crest-lines: sinuous to lunate
- form ratio 15-20 (i.e. shorter and higher)
- 3-D
- lee slope angle <25°
- l 0.6 m - 100's m, h 0.05 m - >10 m
- sediment transport includes rolling, sliding, saltation and intermittent suspension

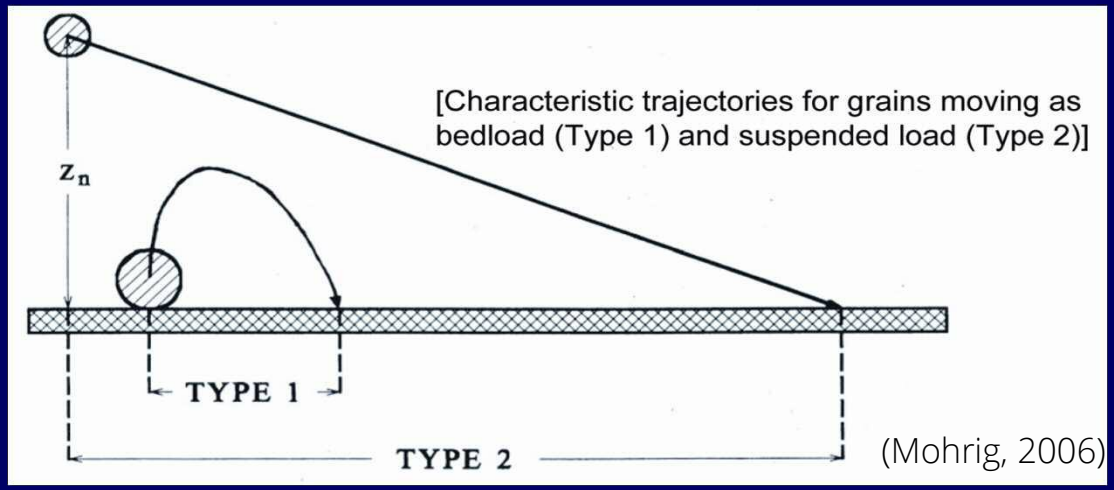
Observation: at higher velocities.....



Some fraction of the transported grains are carried beyond the crest and land on the stoss surface of the next dune downstream

Observation: Transition from dunes to upper plane bed





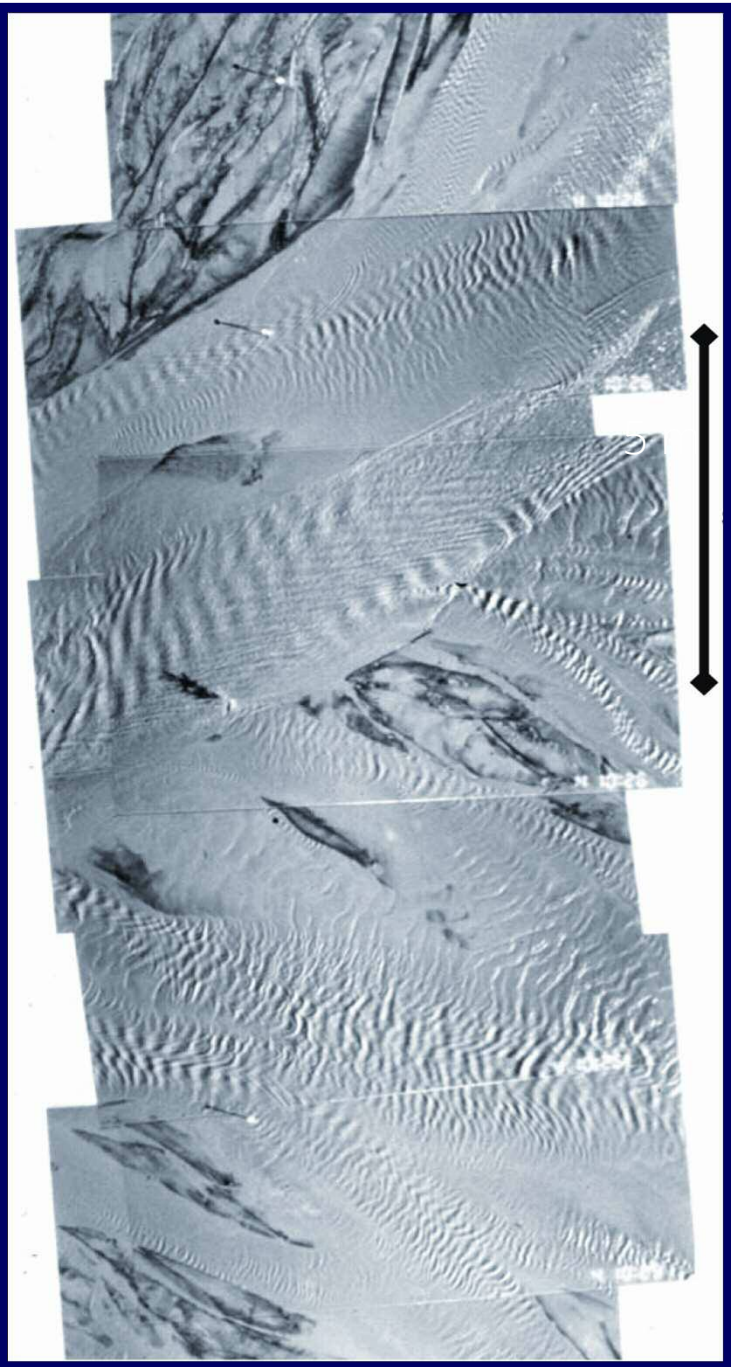
(Mohrig, 2006)

Washout occurs when:

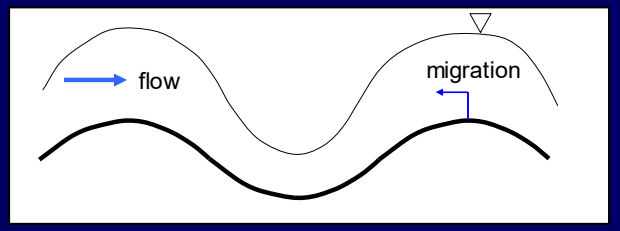
- characteristic excursion lengths of a significant number of size classes become long relative to the distance of the lee face (i.e. suspension is dominant mode of sediment transport)
- since particles bypass lee face they are not incorporated into bedform
- dune crests planed off and height decreases

As flow velocities increase dunes become longer and lower, 'washing out' until finally they disappear entirely, giving way to a planar bed surface over which abundant suspended load as well as bed load is transported close to the bed in a "traction carpet"

Inphase wave bedforms (Antidunes)



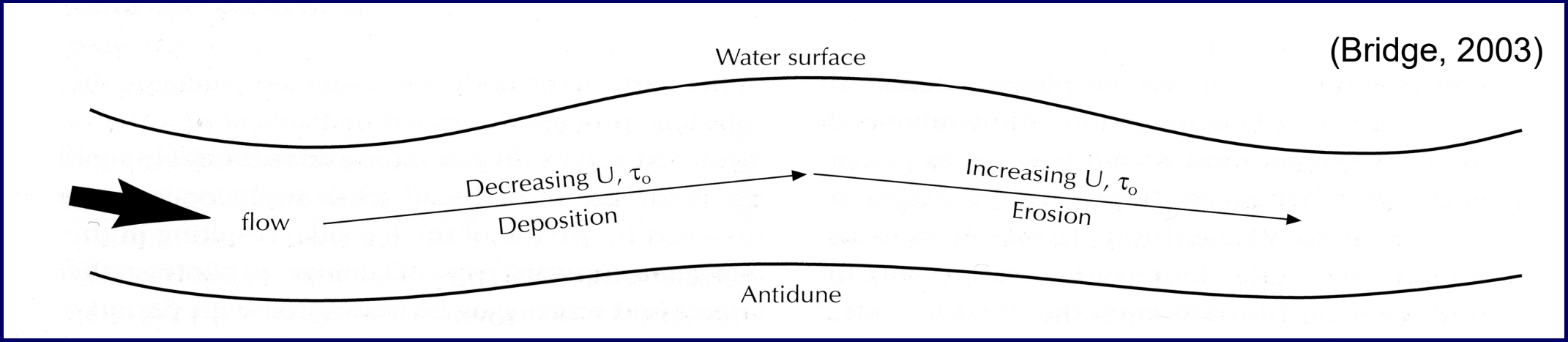
- sinusoidal forms with accompanying in-phase water waves

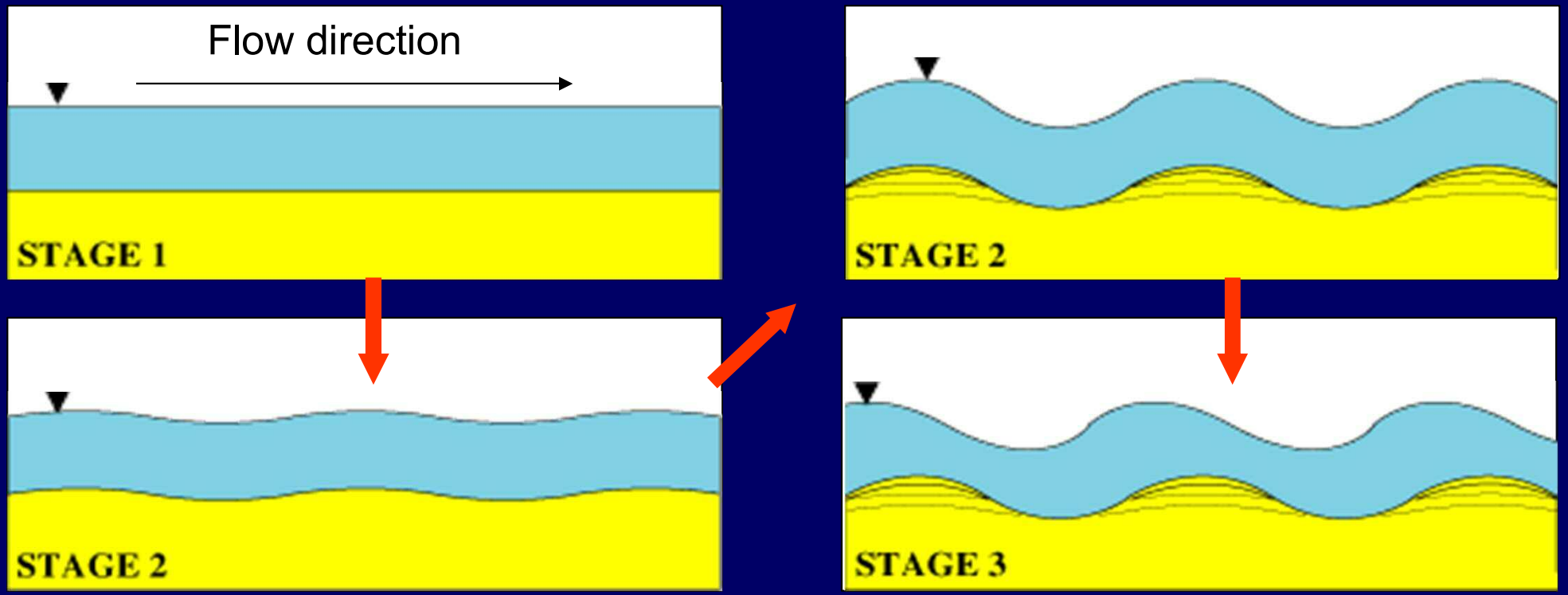


- occur in supercritical flows (i.e. Fr>1) (i.e. for flow depths of 0.1m and 1 m, need mean flow velocities >1 m/s and 3 m/s respectively)
- commonly occur in long trains
- individual waveforms undergo cycles of amplification and destruction

Trains of surface waves indicating the presence of antidunes in braided channels of the tailings basin of the Hibbing Taconite Mine, Minnesota, USA. Flow is from top to bottom. (Mohrig, 2006)

True antidunes (upstream migrating waves) have the following cyclical behaviour





Important points to remember

Lower plane bed only forms in coarse sand or coarser

Ripples form only in medium sand or finer
Ripple spacing: increases with grain size
No clear relationship between ripple size and either flow strength or water depth although planform may vary with flow strength and/or flow duration

Dune spacing: increases with depth
increases with velocity (washing out)

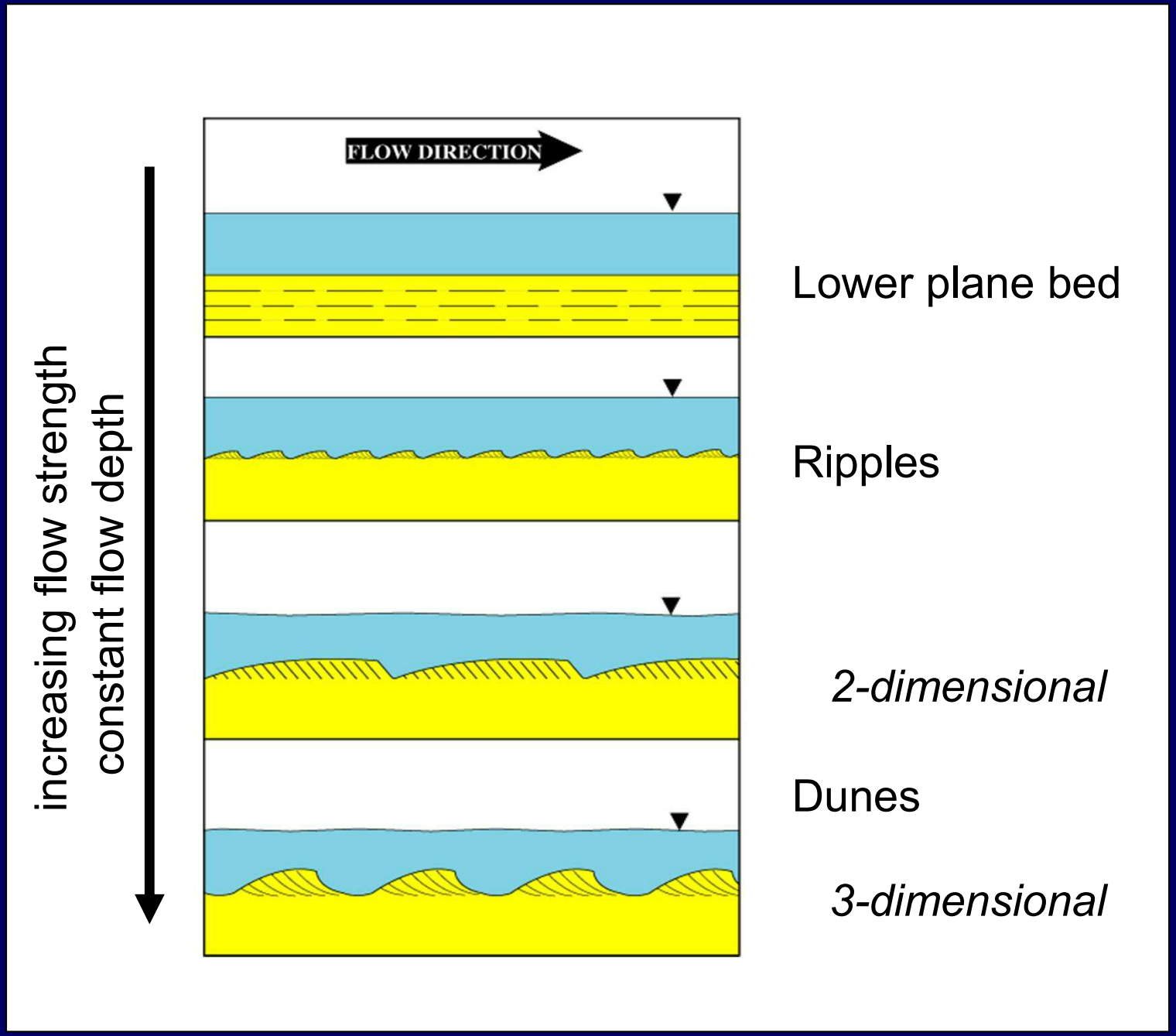
Dune height: increases with depth
increases and then decreases with velocity (washing out)

Upper plane bed will form in any grain size at very high shear stresses where there is a dominance of transport in suspension

Antidunes require a free surface and that the flow is supercritical (i.e. Fr~1)

Sequence of bed forms produced under unidirectional flows

Note:
- fine to coarse sands
- not all bed forms will develop over the fine to coarse sand size range



With even further increase in flow strength.....

